# Introduction

## Smart Analytics for Big Data

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## Grenoble – INP Researcher

- AI as a performance lever of firms and industrial networks
- AI a product or product component (B2B, B2C)

## ACCEL4.0 (2021-24) → ACCEL4.S (2024-27)

- "Accelerating sustainable industry in the region: from research to skills" supported by E. Perret et M-A Le Dain

## Teacher at Industrial Engineering School

- ex. Resp. Supply Chain dept.
- EFELIA (école française de l'IA) – MIAI (institut interdisciplinaire de l'IA) - Iticia (IA AURA)
- AI Learning program and certification in continuous learning

# Introduction

# Introduction

This course is about:

## Big Data

- most cited definition of big data includes the 3Vs (**Volume, Variety, and Velocity**) *Laney (2001)*
- big data should include '**Value**' *(Gantz et al., 2011)*
- big data should also have '**Veracity**' *(Zikopoulos et al., 2013)*

## Data Sciences

- Data Manager
- Data Engineer
- Data Miner
- Data Analyst
- Data Scientist

# Tools

## Data Science, Machine Learning, Statistics, AI, etc

## More than 30 models and families

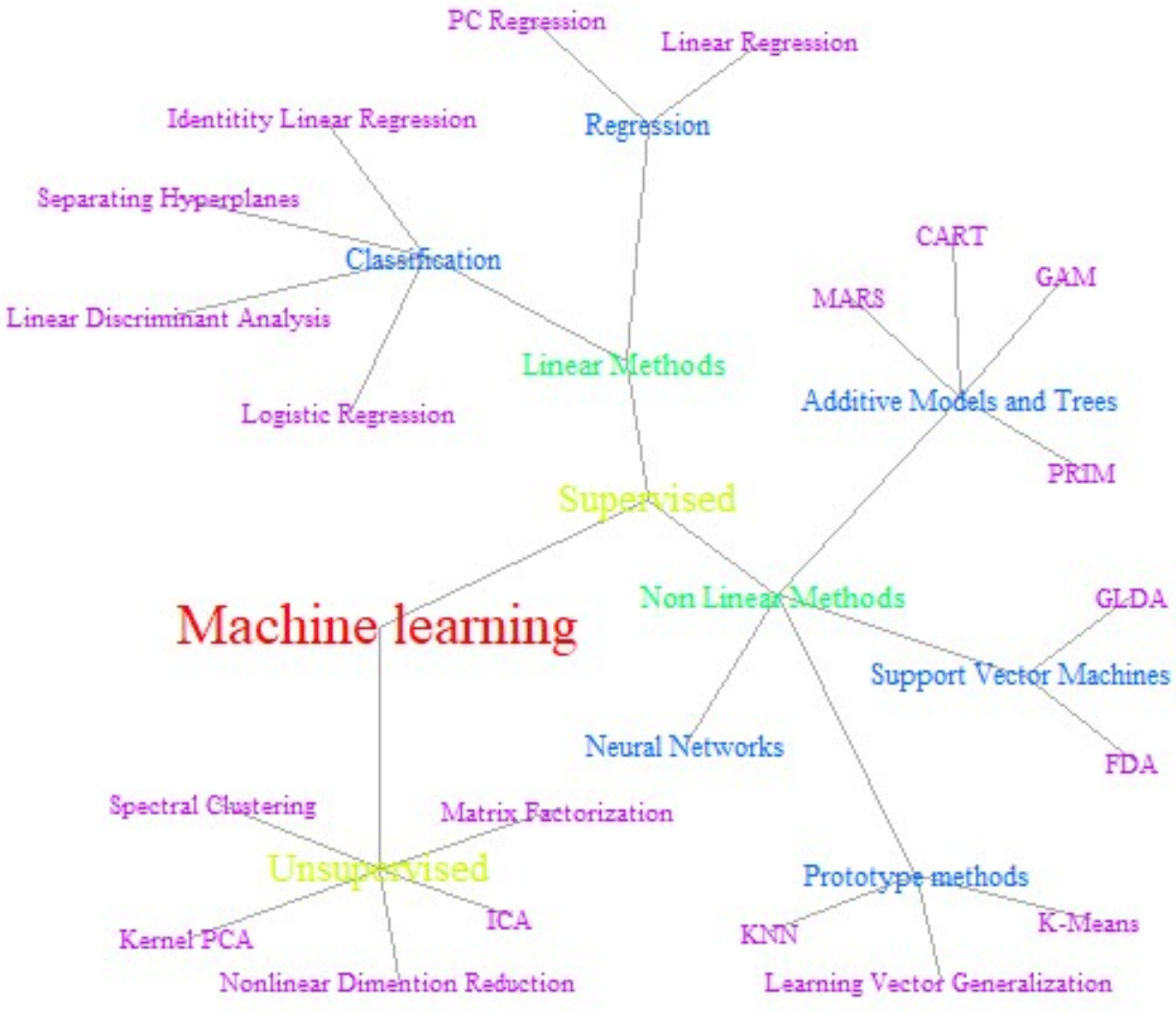
- Generalized Additive Model
- Generalized Linear Model
- Generalized Non-Linear Model
- Multinomial Logistic Regression
- Multinomial Probit
- Linear Model
- Indicator (Identity) Linear Model
- Logistic Regression
- Probability (Unit) Regression
- Cumulative Logistic Regression
- Cumulative Probability (Unit) Regression
- Complementary Log-Log Model
- Continuation-Ratio Logistic Regression
- Hierarchical Logistic Regression
- Poisson Log-Log Regression
- Negative Binomial Regression
- Conditional Regression Model
- Mixed Logistic Regression
- Linear Discriminant Analysis
- Quadratic Discriminant Analysis
- Support Vector Machines
- Order Support Vector Machines
- Support Vector Ordinal Regression
- K Nearest Neighbours
- Classification And Regression Trees
- Separating Hyperplanes
- Neural Networks
- Feed-Forward Neural Networks
- Bayesian Neural Networks
- Markov Chains
- Markov Chains Monte Carlo
- Deep Neural Networks

# Several criterias differenciating the methods

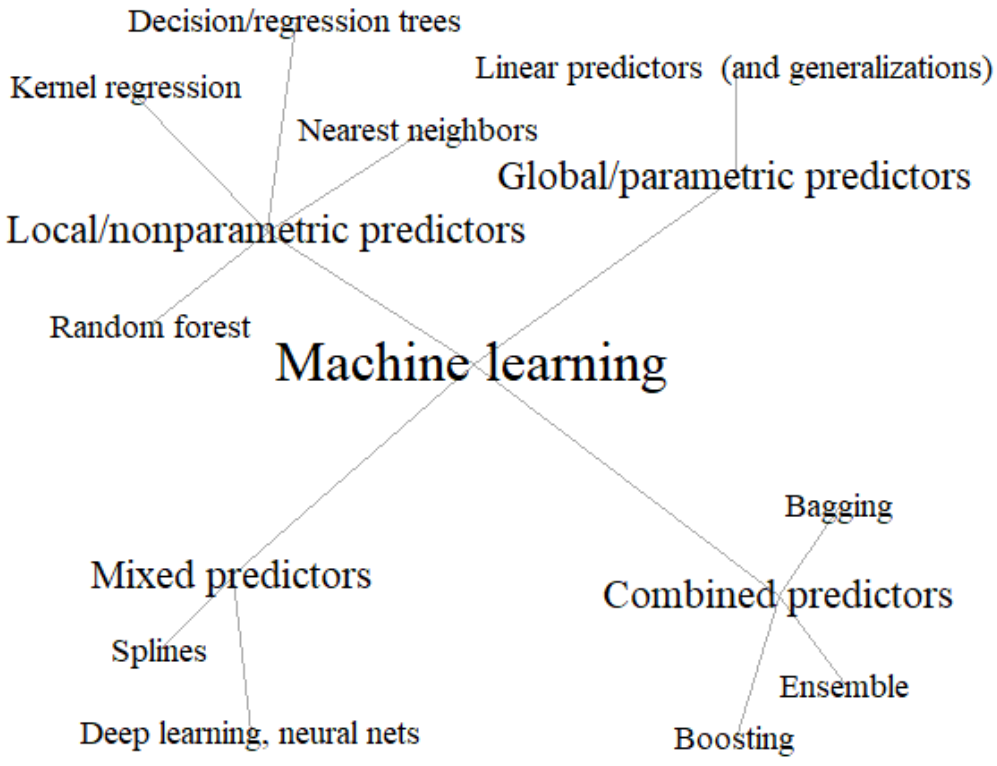
- Supervised
- Unsupervised
- Additive
  - Linear
  - Non-linear
- Generative
- Discriminative
- Parametric
- Semi-parametric
- Non-parametric
- Classification
- Regression
- Binary data
- Multinomial data
- Ordinal data
- Count data
- Continuous data
- Explicative
- Data driven

# This course also is about:

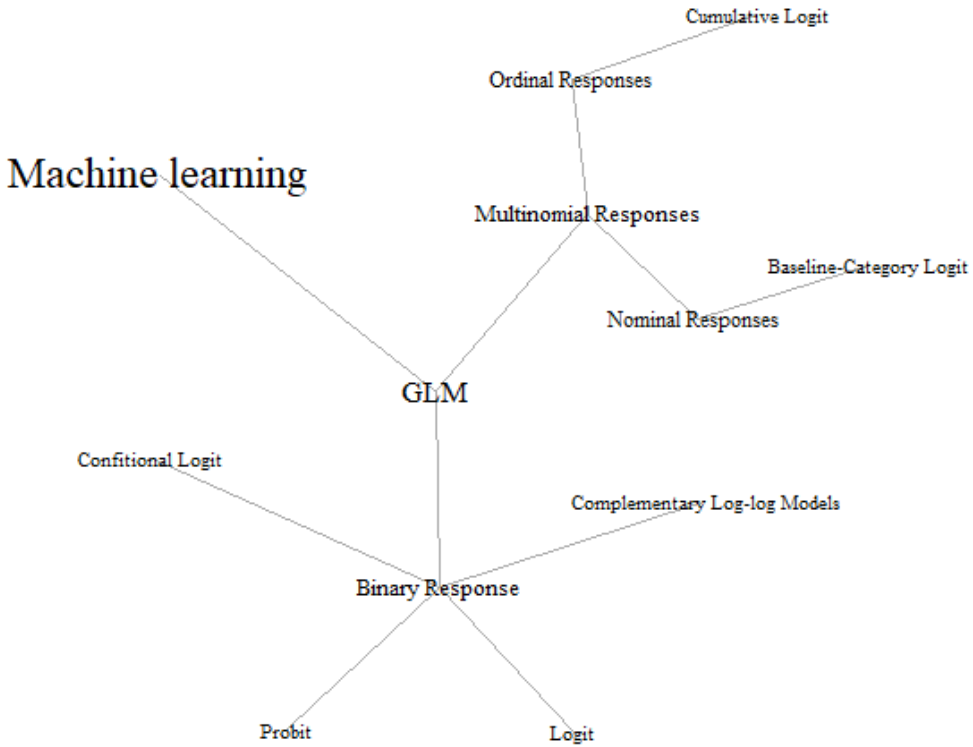
## Hastie et al. (2009) *reference*



# Mullainathan and Spiess (2017) *reference*



# Agresti (2007) *reference*



# Artificial Intelligence (AI)

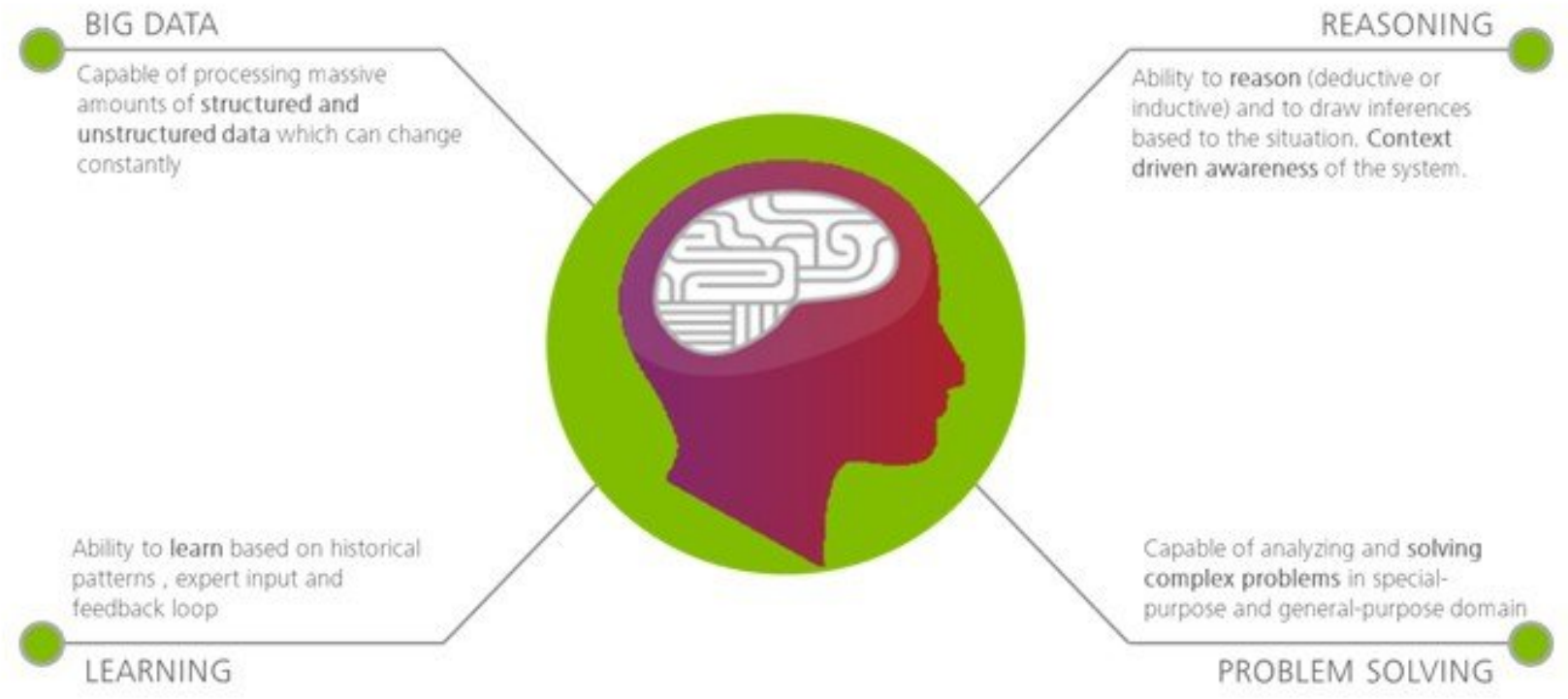



Figure 1: AI schema

# Class organisation

# Goals

*Be DM, DA and DS supervisor in the Big Data context*

# Teachers and Industrial Contributors

- Dr *Genoveva Vargas Solar*, CNRS, HADAS Group
- Nhung Dang, Phd Student, Grenoble INP Génie industriel
- Alexandre Zouaoui, Neovision
- Sébastien Raynoird-Thal, CEO Efimove
- Amélie Favreau and Alexandre Delmotte, lawyers UGA
- *Iragaël Joly* [1], Grenoble INP Génie industriel

[1] corresponding teacher: iragael.joly@grenoble-inp.fr

# Organisation of the course

## Pillars of the course

- Data supply-chain: from data collection and production, storage and organization, management, exploitation and analysis, and communication.
- Big data and dynamic process of analysis needs transparent, repeatable and reproducible technics.
- Backward presentation: from **needs** to solutions

# Applied lessons

## Industrial Data Challenges

- Central Kitchen Menu scoring and optimisation
- Product test and PID optimization
- Equipement and intervention scoring and Predictive maintenance

$\Rightarrow$ Defense (Engineers and Teachers) + Technical report

## Semester project

- Mobility behavior analysis

$\Rightarrow$ Defense (Resaerchers and Teachers) + Scientific Report

# Planning

| | | | 8h-9h30 | 9h45-11h15 | 11h30-13h | | Comments |
|---|---|---|---|---|---|---|---|
| 1 | 22-sept-25 | I. Joly | | AI & Data Science overview | Lifecycle, AI regulation, sustainability, GenAI in industry | | HomeWork Install PARTY |
| 2 | 29-sept.-25 | I. Joly | Data quality & diagnosis | Data quality & diagnosis | | | |
| 3 | 6-oct.-25 | G Vargas | *data management, des pipelines,* | *gouvernance, de la reproductibilité, de la responsabilité et de la durabilité* | ***Intro - TP Mobility*** | | HomeWork Mobility Projet : littérature review |
| 4 | 13-oct.-25 | I. Joly | Algorithms survey - Regression | **Feature selection Dimensionality reduction, variable importance** | | | HomeWork Mobility Projet : theoretical and conceptual modelisation |
| 5 | 20-oct.-25 | I. Joly | Algorithms survey - Classification | Model evaluation Metrics, cross-validation, confusion matrix | Data Challenge introduction | | *HomeWork Data Challenge & Mobility Projet : Data processing & Analytics* |
| | 27-oct.-25 | TOUSSAINT | | | | | |
| 6 | 3-nov.-25 | G Vargas | Workflow & reproducibility | Pipelines, version control, literate programming | | | *HomeWork Data Challenge & Mobility Projet : Data processing & Analytics* |
| 7 | 10-nov.-25 | I. Joly / nhung Dang | Problem definition | *Causality introduction Causal inference basics, DAGs* | TP MOBILITY / TP Data Challenge / Examen - I JOLY | | *HomeWork Data Challenge & Mobility Projet : Analytics & Modeling* |
| 8 | 17-nov.-25 | G Vargas | *federate learning, privacy & collaborative learning* | Industrial seminars **Neovision** | | | *HomeWork Data Challenge & Mobility Projet : Analytics & Modeling* |
| 9 | 24-nov.-25 | I. Joly | Industrial seminars **Efimove** | First presenttion on TP MOBILITY | First presenttion on TP MOBILITY | | *HomeWork Data Challenge & Mobility Projet : Analytics & Modeling* |
| 10 | 1-déc.-25 | I. Joly | Clustering Unsupervised learning, k-means, DBSCAN | Data Challenge | | | |
| 11 | 8-déc.-25 | I. Joly and others | | Data Challenge Evaluation | Data Challenge Evaluation | | |
| | 15-déc.-25 | | | Examen - I JOLY | | | |
| | 22-déc.-25 | | | | | | |
| | 29-déc.-25 | | | | | | |
| 12 | 5-janv.-26 | I. Joly | | Laws and Norms seminars | TP MOBILITY | | |
| 13 | 12-janv.-26 | I. Joly and others | *TP Projet Mobilité - soutenance* | *TP Projet Mobilité - soutenance* | Soutenance ?? | | |
| 14 | 19-janv.-26 | NON ENCADREE ? | TP MOBILITY (sans encadrement) | TP MOBILITY | | | |
| 15 | 26-janv.-26 | | | | | | |
| | | | Rendu rapport 28/01 | | | | |

Figure 2: Planning 2024 - Version sept

# Tools

- RStudio = R + Rmarkdown
- Other tools for DM: Kaggle, Docker, ...

# Evaluation

| Evaluation | Type | Weight |
|---|---|---|
| Individual evaluations | TP, MCQ, **exam** | 30% |
| **Industrial Data Challenge** | Team Work | 30% |
| | Defense | 0.5 |
| | Technical Report | 0.5 |
| **Mobility project** | Team work | 40% |
| | Defense | 0.4 |
| | Scientific Report | 0.6 |

# Plan and Organisation of the lesson

# Today: Reproducible Data Science Project

1. Introduction (15min)
2. Collaborative workshop (2h)
   - Management of an AI project - *Fresque de L'IA*
   - Share, discuss to built a general overview of AI project
   - Team work
3. Conclusion (15 min)

## Out of the class - before next week

1. TODO: Read Definitions
2. TODO: Install Party (15min)
   install R and RStudio
   google: Posit RStudio → follow the instructions
   you'll find links and procedure to install R and RStudio
3. TO READ: Project Management CRISP-DM and Reproducible research (1h)

# Next Week: Project management; Reproducible document; Data quality diagnostic

- *in-class*: introduction & demo
- **Pre requisite** : *out-of-class*: **installing & reading & experimenting**

# References

Agresti, Alan. 2007. *An Introduction to Categorical Data Analysis, Second Edition*.

Hastie, Trevor, Robert Tibshirani, and Jerome Friedman. 2009. *The Elements of Statistical Learning: Data Mining, Inference, and Prediction*. Springer Science & Business Media.

Mullainathan, Sendhil, and Jann Spiess. 2017. "Machine Learning: An Applied Econometric Approach." *Journal of Economic Perspectives* 31 (2): 87–106. https://doi.org/10.1257/jep.31.2.87.